

Lierda FB36 快速上手应用指导

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文件修订历史

文档版本	变更日期	修订人	审核人	变更内容
Rev1.0	24-03-01	ZLJ	SCF	初始版本

适用模块选型

序号	模块型号	支持频段	尺寸(mm)	模组简介
1	L-NLEFB36-G5PP4	2.4GHz ISM Band		

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1 引言

本文主要包括 WIFI、BLE、SLE 这三个模块的基本使用方法，保证用户能够快速上手我司设备，并进行基本功能和验证。

FB36 系列支持 2.4G 频段 WIFI、蓝牙 BLE 5.2 和星闪 SLE 1.0 功能，同时支持三种组网方式。

备注

后面几个章节中涉及的 AT 指令，可参考《Lierda FB36 AT 指令手册》进行操作。

2 WIFI 使用示例

主要分为 AP 模式使用示例和 STA 模式使用示例，以及 ping 和 iperf 打流应用示例。

2.1 AP 模式示例

2.1.1 概述

无线 AP(Access Point, 无线接入点)用于无线网络的无线交换机，也是无线网络的核心，俗称“热点”。FB36 是一块标准无线网卡，通过驱动程序使其提供与 AP 一样的无线信号转接、路由等功能(即：Soft-AP)，其他便携式设备可以通过 802.11 协议与其互联。

2.1.2 AP 使用示例

```
//复位单板
AT+RST
boot.

//设置单板的 MAC 地址
AT+MAC=90:22:33:44:55:66
APP|OK
OK

//开启 AP 模式
AT+STARTAP="XXX",6,2,"123456789"
at_start_softap ssid[XXX]
drv_soc_ioctl ioctl_cmd->cmd=7.
drv_soc_ioctl ioctl_cmd->cmd=9.
drv_soc_ioctl ioctl_cmd->cmd=17.
drv_soc_ioctl ioctl_cmd->cmd=13.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
```

```
drv_soc_ioctl ioctl_cmd->cmd=0.
```

```
drv_soc_ioctl ioctl_cmd->cmd=1.
```

```
drv_soc_ioctl ioctl_cmd->cmd=3.
```

```
OK
```

```
OK
```

```
Srv:547:recive event = 9
```

```
Srv:254:cd is not init
```

```
//设置 IP 地址和网关
```

```
AT+IFCFG=ap0,192.168.49.1,netmask,255.255.255.0,gateway,192.168.49.1
```

```
OK
```

```
//启动 dhcp 服务器
```

```
AT+DHCP=ap0,1
```

```
OK
```

```
OK
```

```
//查看 AP 配置信息
```

```
AT+IFCFG
```

```
ap0      ip:192.168.49.1 netmask:255.255.255.0 gateway:192.168.49.1
```

```
        ip6: FE80::9222:33FF:FE44:5766/64
```

```
        HWaddr 90:22:33:44:57:66 MTU:1500 Running Default Link UP
```

```
lo       ip:127.0.0.1 netmask:255.0.0.0 gateway:127.0.0.1
```

```
        ip6: ::1/64
```

```
        HWaddr 00 MTU:16436 Running Default Link UP
```

```
OK
```

```
OK
```

2.2 STA 模式示例

2.2.1 概述

任何一个接入无线 AP 的设备都可以称为一个站点(STA, Station)。STA 模式即实现与 AP 设备的连接与数据通信。

2.2.2 STA 使用示例

```
//复位单板
AT+RST
boot.
//设置单板的 MAC 地址
AT+MAC=80:22:33:44:55:66
APP|OK
OK

//开启 STA 模式
AT+STARTSTA
EnableSta 1
drv_soc_ioctl ioctl_cmd->cmd=7.
drv_soc_ioctl ioctl_cmd->cmd=9.
drv_soc_ioctl ioctl_cmd->cmd=17.
drv_soc_ioctl ioctl_cmd->cmd=13.
drv_soc_ioctl ioctl_cmd->cmd=35.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=2.
drv_soc_ioctl ioctl_cmd->cmd=41.
OK
```

OK

//扫描可连接网络

AT+SCAN

SCAN

OK

drv_soc_ioctl ioctl_cmd->cmd=14.

hmac_single_hal_device_scan_complete:vap[1] time[771] chan_cnt[13] chan_0[1]
back[0] event[6] mode[0]

Scan::vap[1] find bss_num[32] in regdomain, other bss_num[0]

Srv:547:recive event = 1

Srv:221:cb is not init

//显示扫描结果

AT+SCANRESULT

Srv:1722:sta_scan_results cnt 32

+SCANRESULT:XXX,90:22:33:44:*.**,6,-28,2

+SCANRESULT:lierda-test,0a:69:6c:2f:*.**,1,-36,3

+SCANRESULT:HUAWEI-4004HD,24:6f:8c:ca:*.**,1,-47,2

+SCANRESULT:lierda-acc,12:69:6c:2f:*.**,9,-61,0

+SCANRESULT:lierda-guest,0e:69:6c:25:*.**,9,-65,3

+SCANRESULT:wifitest,c4:04:15:aa:*.**,11,-81,2

OK

//连接 AP，连接成功有 "+NOTICE:CONNECTED" 打印

AT+CONN="XXX",,"123456789"

cmd_sta_connect ssid="XXX"

ConnectToNetwork

Srv:2046: last scantime > 5s, trigger scan

wifi_sta_connect_fill_scan_param:: Specifying an SSID for scanning

OK

OK

drv_soc_ioctl ioctl_cmd->cmd=14.

hmac_single_hal_device_scan_complete:vap[1] time[762] chan_cnt[13] chan_0[1]
back[0] event[6] mode[0]

Scan::vap[1] find bss_num[1] in regdomain, other bss_num[0]

Srv:547:recive event = 1

Srv:216:scan triggered by connect!

Srv:find ssid[XXX] auth type[2] pairwise[3] ft_flag[0]

drv_soc_ioctl ioctl_cmd->cmd=47.

drv_soc_ioctl ioctl_cmd->cmd=47.

drv_soc_ioctl ioctl_cmd->cmd=47.

drv_soc_ioctl ioctl_cmd->cmd=16.

drv_soc_ioctl ioctl_cmd->cmd=47.

drv_soc_ioctl ioctl_cmd->cmd=6.

drv_soc_ioctl ioctl_cmd->cmd=5.

drv_soc_ioctl ioctl_cmd->cmd=6.

drv_soc_ioctl ioctl_cmd->cmd=6.

drv_soc_ioctl ioctl_cmd->cmd=5.

drv_soc_ioctl ioctl_cmd->cmd=1.

drv_soc_ioctl ioctl_cmd->cmd=3.

drv_soc_ioctl ioctl_cmd->cmd=1.

+NOTICE:CONNECTED

drv_soc_ioctl ioctl_cmd->cmd=6.

Srv:547:recive event = 2

Srv:130:cd is not init

//启动 dhcp 客户端

AT+dhcp=wlan0,1

OK

OK

```
//查看 STA 配置信息
```

```
AT+IFCFG
```

```
wlan0      ip:192.168.49.2 netmask:255.255.255.0 gateway:0.0.0.0
            ip6: FE80::8222:33FF:FE44:5566/64
            HWaddr 80:22:33:44:55:66 MTU:1500 Running Default Link UP
lo          ip:127.0.0.1 netmask:255.0.0.0 gateway:127.0.0.1
            ip6: ::1/64
            HWaddr 00 MTU:16436 Running Default Link UP
OK
OK
```

2.3 相关应用示例

测试前准备：设备 1 按照 [2.1.2 AP 使用示例](#) 起 AP，设备 2 按照 [2.2.2 STA 使用示例](#) 起 STA 连接设备 1 的 AP。

2.3.1 Ping 示例

```
//STA 端发起 ping 包
AT+PING=192.168.49.1
OK
[0]Reply from 192.168.49.1: time=20 ms TTL=255

[1]Reply from 192.168.49.1: time<1 ms TTL=255

[2]Reply from 192.168.49.1: time<1 ms TTL=255

[3]Reply from 192.168.49.1: time<1 ms TTL=255

--- 192.168.49.1 ping statistics ---
4 packets transmitted, 4 received, 0 loss

OK
```

2.3.2 Iperf 打流示例

tcp 接收吞吐量:

//先开启 server 监听 (udp 测试需要带 -u 参数)

AT+IPERF=-s,-i,1

APP|AT_IPERF OK

OK

Client 端连接打流后打印:

+IPERF:

Interval	Transfer	Bandwidth
0-1 sec	828 KBytes	6.79 Mbits/sec
1-2 sec	989 KBytes	8.11 Mbits/sec
2-3 sec	1.36 MBytes	11.4 Mbits/sec
3-4 sec	942 KBytes	7.72 Mbits/sec
4-5 sec	964 KBytes	7.90 Mbits/sec
0-5 sec	5.25 MBytes	8.23 Mbits/sec

OK

tcp 发送吞吐量:

//监听开启之后客户端发起连接 (udp 测试需要带 -u 参数)

AT+IPERF=-c,192.168.49.1,-i,1,-t,1

APP|AT_IPERF OK

OK

+IPERF:

Interval	Transfer	Bandwidth
0-1 sec	1.13 MBytes	9.44 Mbits/sec
1-2 sec	896 KBytes	7.34 Mbits/sec
2-3 sec	1.38 MBytes	11.5 Mbits/sec
3-4 sec	896 KBytes	7.34 Mbits/sec
4-5 sec	1.00 MBytes	8.39 Mbits/sec
0-5 sec	5.25 MBytes	8.67 Mbits/sec

OK

3 SLE 使用示例

3.1 概述

SLE(Sparklink Low Energy, 低功耗星闪接入技术), 是星闪联盟设计的一种无线短距离通信技术。

3.2 SLE 使用示例

3.2.1 配对连接

Server 端:

//使能 SLE

AT+SLEENABLE

[ACore] sle enable cbk in, result:0

sle enable

OK

//查询 SLE 地址

AT+SLEGETADDR

[SLE DD SDK] get addr: 00:00:00:00:00:00

OK

//设置地址

AT+SLESETADDR=0,0x012233445566

OK

//设置广播参数

AT+SLESETADVPAR=1,3,200,200,0,012233445566,0,000000000000

[ACore] sle set announce param, handle:1, mode:3, min_interval:c8, max_interval:c8,
tx_power: 0

[ACore] sle set announce param, own addr:0x01:**:**:**:55:66

[ACore] sle set announce param, peer addr:0x00:**:**:**:00:00

OK

//设置广播数据

AT+SLESETADVDATA=1,10,4,aabbccddeeff11223344,11224455

<adv_handle>,<announce_data_len>,<seek_rsp_data_len>,<announce_data>,<seek_rsp_data>

OK

//开启广播（若需要一对多连接，则需要连接后，再发广播进行连接）

AT+SLESTARTADV=1

[ACore] sle start announce in, adv_id:1

OK

[ACore] sle adv cbk in, event:0 status:0

[ACore] sle adv cbk in, event:1 status:0

[ACore] sle adv cbk in, event:2 status:0

[ACore] sle adv cbk in, event:3 status:0

Client 端:

//使能 SLE

AT+SLEENABLE

[ACore] sle enable cbk in, result:0

sle enable

OK

//client 端连接 server

AT+SLECONN=0,0x012233445566

OK

[Connected]

addr:01:**:**:**:55:66, handle:00

//server 端收到 client 连接请求，成功配对连接：

[Connected]

```
addr:ee:**:**:**:8d:5c, handle:00
[ACore] sle adv cbk in, event:7 status:0
```

3.2.2 服务管理

Server 端:

//注册服务端回调函数

AT+SSAPSREGCBK

[AT] ssaps register cbks.

OK

//注册服务端 0x1234: server uuid

AT+SSAPSADDSRV=0x1234

ssaps register server handle (1).

OK

//添加服务

AT+SSAPSADDSERV=0x2222,1

ssaps add_service ret (0).

OK

[AT] ssaps add service cb server_id(1) handle(1) status(0).

//添加属性

AT+SSAPSADDPROPERTY=1,0x2323,3,0xf,2,0x1234

ssaps add property ret (0).

OK

[AT] ssaps add property cb server_id(1) service_handle(1) handle(2) status(0).

//添加属性描述符

AT+SSAPSADDDESCRIPTOR=1,2,0x3333,3,0xf,2,2,0x0100

ssaps add descriptor ret (0).

OK

[AT] ssaps add descriptor cb server_id(1) service_handle(1) property_handle(2)

status(0).

//启动服务

AT+SSAPSSTARTSERV=1

ssaps start_service handle:1, ret:0.

OK

[AT] ssaps start service cb server_id(1) handle(1) status(0).

Client 端:

//注册客户端回调函数

AT+SSAPCREGCBK

[ssap client] client register cbk

OK

//发现服务

AT+SSAPCFNDSTRU=0,0,1,0x1234,0,0xff

OK

[13:56:54.260]收←◆discovery character cbk complete in

[ssap client] find structure cbk client: 0 conn_id:0 status: 0

[ssap client] find structure structure start_hdl:[0x01], end_hdl:[0x02], uuid len:2

[ssap client] structure uuid:[0x22][0x22]

[ssap client] find structure complete cbk client: 0 conn_id:0 status:0, type:1

3.2.3 数据交互

Server→Client:

//Server 通过 handle 发送通知

AT+SSAPSSNDNTFY=0,1,0,2,0x0200

update ssap send report handle: pre handle:ffff, current:0

ssaps notify indicate ret:0.

OK

```
//Client 接收打印
[ssap client] notification info cbk client 0,handle:1, type:0, status:0
-----data-----
02 00
-----

//Server 通过 uuid 发送通知
AT+SSAPSNIFYBYUUID=0,0x2222,0,0xffff,0,2,0x0200
ssaps notify indicate by uuid ret:0.
OK

//Client 接收打印
[ssap client] notification info cbk client 0,handle:1, type:0, status:0
-----data-----
02 00
```

Client→Server:

```
//Client 通过 handle 写数据
AT+SSAPCWRITECMD=0,0,2,0,2,0x8899
OK

//Server 接收打印
[AT] ssaps write request cb server_id(1) conn id(0) handle(2) status(0).
[AT] ssaps write request cb request_id(0) type(0) need_rsp(0) need_authorize(0).
[AT] ssaps write request cb data len(2).
-----data-----
88 99
-----
```

4 BLE 使用示例

4.1 概述

BLE(Bluetooth Low Energy, 低功耗蓝牙), 是蓝牙技术联盟设计的一种个人局域网技术。相较经典蓝牙, 低功耗蓝牙旨在保持同等通信范围的同时显著降低功耗和成本。

下文描述的 BLE 使用示例中, 包括两种连接形式:

- (1) BLE 单板连接手机, 然后进行数传。
- (2) 两块 BLE 单板间进行连接通信。

4.2 BLE 使用示例

4.2.1 单板连接手机

```
//单板使能 BLE
AT+BLEENABLE
[ACore] ble enable cbk in, event:a
OK

//设置地址
AT+BLESETADDR=0,0x112233445566
OK

//设置名称
AT+BLESETNAME=9,jokerbill
OK

//设置广播数据
AT+BLESETADVDATA=0,0x112233445566,0,0,1
OK

//设置广播参数
```

```
AT+BLESETADVPAR=0x32,0x32,0,0x000000000000,0,0x000000000000,7,0,1,0,0x1
```

```
[ACore] ble set adv param min_interval:0x32, max interval:0x32, adv_type:0,
duration:0
```

```
[ACore] ble set adv param, own addr:0x00:**:**:00:00
```

```
[ACore] ble set adv param, peer addr:0x00:**:**:00:00
```

```
OK
```

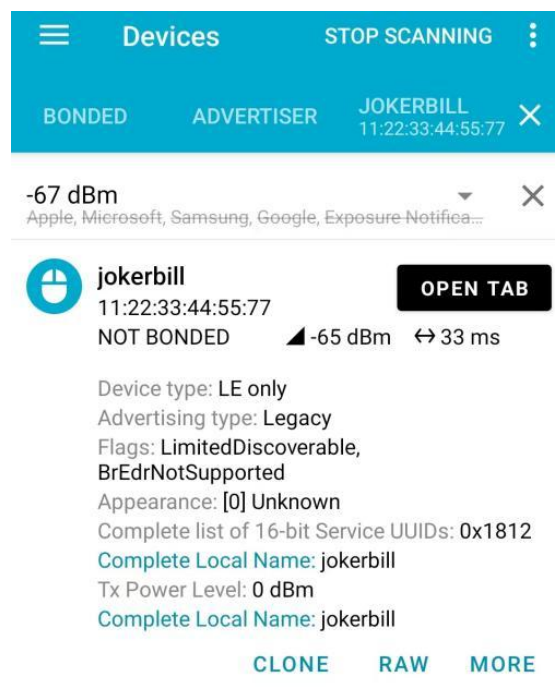
//开启广播

```
AT+BLESTARTADV=1
```

```
[ACore] gap ble start adv in, adv_id:1
```

```
OK
```

手机搜索并连接单板：



连接成功打印：

```
[Connected] addr:4a:**:**:ec:53, handle:00
```

```
[ACore] ble gap adv set terminated cbk event:8
```

4.2.2 两块单板间连接通信

server 端：

//使能 BLE

AT+BLEENABLE

[ACore] ble enable cbk in, event:a

OK

//设置地址

AT+BLESETADDR=0,0x112233445566

OK

//设置名称

AT+BLESETNAME=9,jokerbill

OK

//设置广播数据

AT+BLESETADVDATA=0,0x112233445566,0,0,1

OK

//设置广播参数

AT+BLESETADVPAR=0x32,0x32,0,0x000000000000,0,0x000000000000,7,0,1,0,0x1

[ACore] ble set adv param min_interval:0x32, max interval:0x32, adv_type:0,
duration:0

[ACore] ble set adv param, own addr:0x00:**:**:**:00:00

[ACore] ble set adv param, peer addr:0x00:**:**:**:00:00

OK

//注册 GAP 回调

AT+BLEGAPREGCBK

OK

//注册 server 回调

AT+GATTSREGCBK

OK

//创建 GATT server

AT+GATTSREGSRV=0x1122

[GAP service register][device name] handle: 0x0006, value handle: 0x0007

[GAP service register][device appearance] handle: 0x0008, value handle: 0x0009

[GAP service register][prefer conn param] handle: 0x000a, value handle: 0x000b

[GATTServer]register server: 1

OK

[GATTServer]ServiceStartCallback server: 1 srv_hdl: 1 status: 0

[GATTServer]ServiceStartCallback server: 1 srv_hdl: 5 status: 0

//添加 GATT 服务

AT+GATTSADDSERV=1,0x1812,1

OK

[GATTServer]ServiceAddCallback server: 1 srv_handle: 14 uuid_len: 2 uuid: 1812
status:0

//添加特征

AT+GATTSADDCHAR=1,14,0x2a4a,0x03,0x06,4,01010003

OK

[GATTServer]CharacteristicAddCallback server: 1 srv_hdl: 14 char_hdl: 15
char_val_hdl:16 uuid_len: 2 uuid: 2a4a
status:0

//添加特征描述符

AT+GATTSADDDESCR=1,14,0x2a4a,0x03,2,0100

OK

[GATTServer]DescriptorAddCallback server: 1 srv_hdl: 14 desc_hdl: 17 uuid_len:2,
uuid: 2a4a
status:0

//启动指定服务

```
AT+GATTSTARTSERV=1,14
```

```
OK
```

```
[GATTServer]ServiceStartCallback server: 1 srv_hdl: 14 status: 0
```

```
//开启广播
```

```
AT+BLESTARTADV=1
```

```
[ACore] gap ble start adv in, adv_id:1
```

```
OK
```

```
AdvEnableCallback adv_id: 1, start success
```

Client 端:

```
//使能 BLE
```

```
AT+BLEENABLE
```

```
[ACore] ble enable cbk in, event:b
```

```
OK
```

```
//注册 GAP 回调
```

```
AT+BLEGAPREGCBK
```

```
OK
```

```
//设置地址
```

```
AT+BLESETADDR=0,0x0000000000002
```

```
OK
```

```
//注册客户端回调函数
```

```
AT+GATTCREGCBK
```

```
OK
```

```
//创建 GATT client
```

AT+GATTCREG=0x1234

[GATTClient] client: 1

OK

//连接指定 server

AT+BLECONN=0,0x112233445566

OK

[Connected] addr:11:**.**.**:22:66, handle:00

conn_id: 0 conn_state: 1 pair_status: 1 addr->type 0 disc_reason 0

[GATTClient]Mtu changed----client:1 conn_id:0 mtu size:23

status:0

//server 端收到连接请求，连接成功：

[Connected] addr:00:.**.**:00:02, handle:00**

[ACore] ble gap adv set terminated cbk event:8

conn_id: 0 conn_state: 1 pair_status: 1 addr->type 0 disc_reason 0

[GATTServer]ExchangeMtuReqReceiveCallback--server_id:1 conn_id:0

req_mtu_size:23

status:0

AdvTerminateCallback adv_id: 1 status: 0

//发现服务

AT+GATTCFNDSEV=1,0,0x1812

[GATTClient]Discovery service----client:1 conn_id:0

start handle:14 end handle:17 uuid_len:2

uuid:1812

status:0

OK

```
[GATTClient]Discovery service complete----client:1 conn_id:0 uuid len:2  
uuid:1812status:0
```

//交换 MTU

```
AT+GATTCEXCHMTU=1,0,516
```

OK

```
[GATTClient]Mtu changed----client:1 conn_id:0 mtu size:516  
status:0
```

//Server 端收到交换 MTU 请求：

```
[GATTServer]ExchangeMtuReqReceiveCallback--server_id:1 conn_id:0  
req_mtu_size:516
```

Client→Server:

//Client 通过 handle 写数据

```
AT+GATTCWRITECMD=1,0,16,4,0x12341234
```

OK

```
[GATTClient]Write result----client:1 conn_id:0 handle:16  
status:0
```

//Server 收到 client 写入数据

```
[GATTServer]WriteReqReceiveCallback--server_id:1 conn_id:0  
request_id:2 handle:16 offset:0 need_authorize:0 need_rsp:0 is_prep:0  
data_len:4 data:  
12 34 12 34  
status:0
```

//Client 通过 handle 读数据

```
AT+GATTCREADBYHDL=1,0,16
```

OK

```
[GATTClient]Read result----client:1 conn_id:0  
handle:16 data_len:4
```

```
data:12 34 12 34
```

```
status:0
```

```
//Server 收到 client 读请求
```

```
[GATTServer]ReadReqReceiveCallback--server_id:1 conn_id:0
```

```
request_id:3 handle:16 offset:0 need_authorize:0 need_rsp:1 is_long:0
```

```
status:0
```